Euclid's Elements — Book XI
Proposition 12

Formal Reconstruction — Technical Documentation

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To set up a straight line at right angles to a given plane from a given point in it.

1 Introduction

Proposition XI.12 is the companion construction to XI.11. Proposition XI.11 constructs a perpendicular to a plane from a point outside the plane; XI.12 starts with a point A already lying in the reference plane π and constructs a line through A perpendicular to π.

The classical proof is exceptionally short. Euclid chooses an elevated point, uses XI.11 to obtain one perpendicular to the reference plane, draws through A a parallel to that perpendicular by I.31, and then uses XI.8 to transfer the line–plane perpendicularity to the new line.

The Lean reconstruction follows this classical route, but two incidence obligations that are invisible in the printed proof become explicit. First, the perpendicular obtained from XI.11 may already meet the reference plane at A. In that case it is itself the required line, and there is no new parallel to construct through A. Second, XI.8 returns the foot of the new perpendicular existentially; Lean must prove that this foot is the already specified point A.

These two obligations are handled by the proposition-local theorem `hilbert_XI12_perpendicular_foot_unique`. The other proposition-local helper, `hilbert_XI12_parallel_through_point_in_plane`, isolates the planar I.31 construction and packages its result as ambient spatial parallelism.

2 Euclid's Statement

Euclid's Proposition XI.12 states:

> To set up a straight line at right angles to a given plane from a given point in it.

Let A be the given point of the reference plane π. The required construction is a straight line through A satisfying

$$AD \perp \pi.$$

3 Classical Configuration

Let B be an arbitrary elevated point. By XI.11 draw BC perpendicular to the reference plane, with $C \in \pi$. Through the given point $A \in \pi$, draw AD parallel to BC.

The generic classical configuration is shown in Figure 1. The two blue lines leave the reference plane. Their hidden continuations below the plane are dashed, following the standing Book XI figure convention.

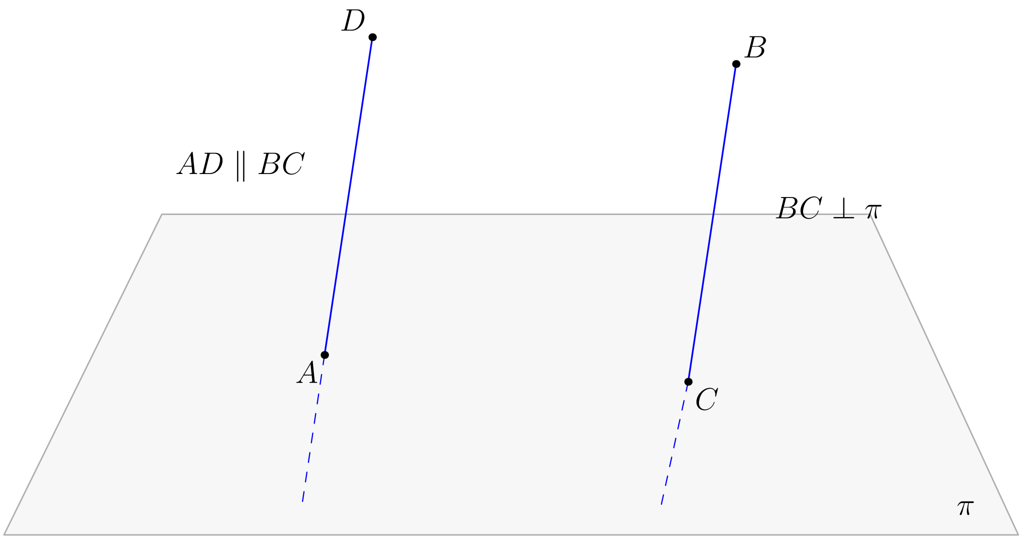

Figure 1: The classical configuration of Proposition XI.12. The points A, C lie in the reference plane π. XI.11 gives $BC \perp \pi$, and I.31 draws $AD \parallel BC$ through A. XI.8 then gives $AD \perp \pi$. The figure shows the generic case $A \neq C$; if $A = C$, the line BC already solves the construction.

4 Classical Proof

Let A be the given point in the reference plane. Choose any elevated point B. By Proposition XI.11 draw

$$BC \perp \pi.$$

Through A, draw

$$AD \parallel BC$$

by Proposition I.31. Since AD and BC are parallel and BC is perpendicular to the reference plane, Proposition XI.8 gives

$$AD \perp \pi.$$

Thus AD is the required straight line.

The local classical dependency chain is therefore

$$\begin{gathered} A \in \pi \\ \Downarrow \\ \text{choose an elevated point } B \\ \Downarrow \\ \text{XI.11} \\ \Downarrow \\ BC \perp \pi \\ \Downarrow \\ \text{I.31} \\ \Downarrow \\ AD \parallel BC, \quad A \in AD \\ \Downarrow \\ \text{XI.8} \\ \Downarrow \\ AD \perp \pi. \end{gathered}$$

Neither XI.9 nor XI.10 occurs in this local proof. The only earlier Book XI propositions used explicitly are XI.11 and XI.8.

5 What is genuinely three-dimensional here?

The I.31 step is planar, but the input line BC is not a line of the reference plane: it is perpendicular to that plane and meets it only at its foot C. To apply a planar parallel theorem through A, the formal reconstruction first has to place the spatial line and the point A in a common auxiliary plane.

In the nontrivial branch this plane is

$$\sigma = \mathrm{plane}(BC, A).$$

Inside PlaneGeo(σ), I.31 becomes an ordinary planar construction. Its output must then be transported back to ambient three-space and packaged as

$$\mathrm{HilbertSpaceLinesParallel}(BC, AD).$$

Only after this packaging can XI.8 be applied.

The other genuinely spatial ingredient is the meaning of a line perpendicular to a plane. The predicate

```
HilbertLinePerpendicularPlaneAt Geo l pi A
```

contains the incidence of the foot with both line and plane and the universal assertion that the line is perpendicular to every line of the plane through the foot. This universal structure is what makes the foot-uniqueness argument possible.

6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_12
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (pi : S.Plane)
    (A : Geo.Point)
    (hApi : S.OnPlane A pi) :
    exists l : Geo.Line,
      HilbertLinePerpendicularPlaneAt Geo l pi A
```

The theorem therefore returns an ambient line l that is perpendicular to the given plane exactly at the prescribed point A.

7 Spatial / Planar Architecture Used

7.1 Ambient spatial layer

The ambient proof uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt
```

As throughout the current Book XI architecture, there is no global ambient

```
HilbertOrder Geo
HilbertCongruence Geo
```

installed to make plane geometry available in space.

7.2 Induced planar geometry

The only new plane slice constructed specifically by XI.12 is σ, the plane through the XI.11 perpendicular and the point A, in the branch where A does not already lie on that line.

The I.31 construction is then carried out in

$$\mathrm{PlaneGeo}(\sigma).$$

The helper `hilbert_XI12_parallel_through_point_in_plane` selects two points of the ambient line, views them together with A as `PlanePoint`s of σ, applies the established planar parallel-existence theorem, and converts the result back to an ambient line.

7.3 Plane–space bridges

The principal bridge operations used locally are

```
planeGeo_primCollinear_to_ambient
hilbert_on_line_of_primCollinear_with_two_on_line
hilbert_mem_pointLine_iff_onLine
HilbertSpaceIncidence.line_in_plane
```

The planar I.31 output is not identified with ambient spatial parallelism by definition. The proof explicitly establishes ambient disjointness and supplies σ as the common carrier plane.

8 Proof Architecture of the Reconstruction

8.1 1. Choosing an elevated point and invoking XI.11

The proof first calls

```
hilbert_point_off_plane
```

to choose a point $B \notin \pi$. It then invokes

```
euclid_proposition_11_11
```

and obtains an ambient line l and a point F such that

$$B \in l, \qquad l \perp \pi \text{ at } F.$$

At this point the formal proof has exactly the spatial object that Euclid denotes by BC.

8.2 2. The exceptional incidence branch

The classical text immediately asks for a parallel through A. Formally, however, one must first decide whether A already lies on the XI.11 line l:

```
by_cases hAl : H.OnLine A l
```

If $A \in l$, then both A and the perpendicular foot F lie in the reference plane and on l. The helper

```
hilbert_XI12_perpendicular_foot_unique
```

proves

$$A = F.$$

Hence $l \perp \pi$ at A, and the theorem is complete in this branch. No I.31 construction is needed.

8.3 3. Why the perpendicular foot is unique

The helper has the normalized signature

```
theorem hilbert_XI12_perpendicular_foot_unique
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (pi : S.Plane)
    (l : Geo.Line)
    (C A : Geo.Point)
    (hPerp :
      HilbertLinePerpendicularPlaneAt Geo l pi C)
    (hAl : H.OnLine A l)
    (hApi : S.OnPlane A pi) :
    A = C
```

Suppose $A \neq C$. Since the two distinct points A, C lie both on l and in π, the spatial incidence axiom `line_in_plane` forces

$$l \subset \pi.$$

But $l \perp \pi$ at C. The universal defining clause of line–plane perpendicularity can therefore be applied to l itself, giving

$$l \perp l.$$

The established theorem `hilbert_linesPerpendicularAt_ne` then yields $l \neq l$, a contradiction. Thus the common point of a perpendicular line and its plane is unique.

8.4 4. Constructing the auxiliary carrier plane

In the second branch $A \notin l$. The theorem

```
hilbert_plane_through_line_and_external_point
```

constructs a plane σ satisfying

$$l \subset \sigma, \qquad A \in \sigma.$$

This is the precise formal replacement for the plane that the classical diagram leaves implicit when it draws a line through A parallel to the spatial line BC.

8.5 5. I.31 inside the induced plane

The helper

```
hilbert_XI12_parallel_through_point_in_plane
```

has the normalized signature

```
theorem hilbert_XI12_parallel_through_point_in_plane
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (sigma : S.Plane)
    (l : Geo.Line)
    (A : Geo.Point)
    (hlsigma : HilbertLineInPlane Geo l sigma)
    (hAsigma : S.OnPlane A sigma)
    (hAl : Not (H.OnLine A l)) :
    exists m : Geo.Line,
      H.OnLine A m /\
      HilbertSpaceLinesParallel Geo l m
```

The proof chooses two distinct points B, C on l, proves that B, C, A are noncollinear in PlaneGeo(σ), and then calls

```
hilbert_parallel_through_point_exists
```

inside the induced plane.

This is the formal I.31 step. The production module imports `Proposition31`, but the local helper calls the established lower-level parallel-existence theorem directly.

8.6 6. Packaging planar parallelism as spatial parallelism

The planar theorem initially gives disjoint `PointLine` carriers in PlaneGeo(σ). The proof constructs the actual ambient carrier m, proves that it lies in σ, and shows that an ambient intersection point of l and m would contradict the planar disjointness.

It can therefore package

$$l \parallel m$$

in the spatial sense

```
HilbertSpaceLinesParallel Geo l m
```

with σ as the explicit common carrier plane. This step excludes skew lines by construction.

8.7 7. XI.8 transfers perpendicularity

The proof now has

$$l \parallel m, \qquad l \perp \pi.$$

It applies

```
euclid_proposition_11_8
```

and obtains a point D such that

$$m \perp \pi \text{ at } D.$$

This is the exact classical XI.8 step. The only extra issue is that the formal theorem returns D existentially, whereas the target of XI.12 requires the foot to be the prescribed point A.

8.8 8. The second foot normalization

The constructed line m passes through A, and the original hypothesis gives $A \in \pi$. Applying `hilbert_XI12_perpendicular_foot_unique` to the XI.8 conclusion therefore gives

$$A = D.$$

After this substitution the conclusion is exactly

$$m \perp \pi \text{ at } A.$$

This closes the nontrivial branch and completes XI.12.

9 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_11
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition31
```

The two proposition-local declarations introduced by XI.12 are

```
hilbert_XI12_perpendicular_foot_unique
hilbert_XI12_parallel_through_point_in_plane
```

and the public theorem is

```
euclid_proposition_11_12
```

The first helper is a general uniqueness fact about a perpendicular line and its plane. The second is a general plane-slice packaging fact for parallel construction. Both are currently located in the proposition module; their signatures are sufficiently general that later promotion to `Hilbert3DInterface` may be considered if subsequent Book XI proofs reuse them.

10 Hidden Formal Data

10.1 The classical parallel construction has an exceptional case

Euclid chooses an arbitrary elevated point B, drops the XI.11 perpendicular, and then draws through A a parallel to it. The printed proof does not discuss the possibility that the foot of the XI.11 perpendicular is already A.

The formal proof cannot ignore this case. If $A \in l$, foot uniqueness shows that A is the foot and l is already the required perpendicular. Only when $A \notin l$ is the I.31 construction invoked.

10.2 I.31 needs an explicit plane slice

A planar parallel theorem cannot be applied directly to an arbitrary ambient spatial line and a point. The carrier plane σ through l and A is constructed first, and the entire I.31 step is performed inside PlaneGeo(σ).

10.3 Planar and spatial parallelism are different data

The I.31 result is initially a planar point-pair parallel relation. XI.8 expects `HilbertSpaceLinesParallel`, which includes an explicit common plane and ambient disjointness. The helper proves and packages these data rather than treating the two relations as definitionally identical.

10.4 XI.8 returns a new foot

XI.8 proves that the parallel line is perpendicular to π, but its formal conclusion contains an existentially returned foot D. The equality $D = A$ is a genuine incidence theorem and is discharged by the same foot-uniqueness helper used in the first branch.

11 Relationship with Earlier Book XI Propositions

The classical local Book XI dependencies are exactly XI.11 and XI.8.

XI.11 supplies a line perpendicular to the reference plane from an elevated point:

$$B \notin \pi \Longrightarrow \exists l, F, \quad B \in l, \quad l \perp \pi \text{ at } F.$$

XI.8 supplies the transfer along spatial parallel lines:

$$l \parallel m, \quad l \perp \pi \Longrightarrow m \perp \pi.$$

The production file imports both propositions directly. XI.9 and XI.10 are neither classical local dependencies nor Lean dependencies of XI.12.

XI.4 is not called directly by XI.12, but it remains in the transitive proof history through XI.11 and XI.8.

12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The only explicit classical Book I proposition in XI.12 is I.31. In the formal proof its role is implemented by

```
hilbert_parallel_through_point_exists
```

inside PlaneGeo(σ).

The XI.12-specific parallel helper does not itself require `HilbertSpaceEuclidean`; its theorem signature stops at spatial order and congruence. The Euclidean Group IV assumption of the public XI.12 theorem is inherited from the calls to XI.11 and XI.8.

Book Zero contributes no new XI.12 declaration. The final public axiom profile contains the already accepted inherited dependency `bookZero_nullSegment1`, whereas both new proposition-local helpers audit without that dependency.

The ambient three-space continues to obey the Book XI instance discipline: ordinary order and congruence are used only inside explicit `PlaneGeo` slices, while the ambient layer uses the spatial classes.

13 Axiomatic Status

The audit command for the public theorem is

```
import CGJteamLab.Proposition11_12

#print axioms Geometry.euclid_proposition_11_12
```

and Lean reports

```
'Geometry.euclid_proposition_11_12' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The two new helpers were audited separately:

```
'Geometry.hilbert_XI12_perpendicular_foot_unique'
depends on axioms: [propext, Classical.choice, Quot.sound]

'Geometry.hilbert_XI12_parallel_through_point_in_plane'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

Thus XI.12 introduces no new proposition-specific global axiom constant. In particular, the two new helpers do not introduce `bookZero_nullSegment1`; that dependency is inherited by the public theorem through the established construction chain.

The explicit signature nevertheless contains

```
[HSE : HilbertSpaceEuclidean Geo]
```

so Hilbert Group IV is a genuine hypothesis of the current production theorem. It is consumed through XI.11 and XI.8. No continuity assumption is used.

The current theorem records the strength of the Euclid-faithful proof route. It should not be read as a minimality claim about the abstract existence theorem.

At this checkpoint:

- new proposition-specific geometric axiom: no;
- new spatial axiom declaration: no;
- Euclidean parallel axiom / Hilbert Group IV: used, via XI.11 and XI.8;
- new continuity assumption: no;
- new proposition-local helper theorem: yes, two;
- new reusable `Hilbert3DInterface` theorem: no;
- new `Hilbert3DAxioms` declaration: no;

- new `PlaneGeo` bridge theorem: no;
- global ambient `HilbertOrderGeo`: not installed;
- global ambient `HilbertCongruenceGeo`: not installed;
- public theorem compiles: yes, as reported at this checkpoint;
- full repository build: clean, as reported at this checkpoint;
- axiom audit: completed;
- `sorry`/`admit`: no.

14 Dependency Summary

The classical local chain is

$$\begin{gathered} A \in \pi \\ \downarrow \\ B \text{ elevated} \\ \downarrow \\ \text{XI.11} \\ \downarrow \\ BC \perp \pi \\ \downarrow \\ \text{I.31} \\ \downarrow \\ AD \parallel BC \\ \downarrow \\ \text{XI.8} \\ \downarrow \\ AD \perp \pi. \end{gathered}$$

The actual formal chain is

$$
\begin{array}{c}
A \in \pi \\
\downarrow \\
\texttt{hilbert\_point\_off\_plane} \\
\downarrow \\
B \notin \pi \\
\downarrow \\
\text{XI.11} : l \perp \pi \text{ at } F \\
\downarrow \\
\texttt{by\_cases}\ \ A \in l \\
\downarrow
\end{array}
$$

$$
\begin{array}{c|c}
A \in l & A \notin l \\
\downarrow & \downarrow \\
A = F & \sigma = \text{plane}(l, A) \\
\downarrow & \downarrow \\
l \perp \pi \text{ at } A & \text{I.31 in PlaneGeo}(\sigma) \\
 & \downarrow \\
 & \texttt{HilbertSpaceLinesParallel}\ l\, m \\
 & \downarrow \\
 & \text{XI.8} : m \perp \pi \text{ at } D \\
 & \downarrow \\
 & D = A
\end{array}
$$

$$
\begin{array}{c}
\downarrow \\
\texttt{euclid\_proposition\_11\_12}.
\end{array}
$$

The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	yes, via XI.11 and XI.8
Direct production imports	XI.11, XI.8, I.31 module
Classical explicit Book XI dependencies	XI.11, XI.8
XI.9 / XI.10	not used
New proposition-local helper theorem	yes, two
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
`sorry/admit`	no

15 Conclusion

Proposition XI.12 has a minimal classical proof but a revealing formal shape. Euclid's three-step route remains visible:

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8}.$$

Lean adds no alternative geometry. It makes explicit the carrier plane needed for the planar parallel construction, the exceptional case in which the XI.11 perpendicular already passes through the prescribed point, the conversion from planar to spatial parallelism, and the identification of the foot returned by XI.8.

The two new helpers are axiomatically clean relative to the inherited Book Zero dependency. The public theorem has the same global axiom profile as XI.11 and XI.8:

$$[\texttt{propext}, \texttt{Classical.choice}, \texttt{Geometry.bookZero\_nullSegment1}, \texttt{Quot.sound}].$$

The current result remains fully synthetic and uses no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure. Its Group IV hypothesis belongs to the chosen Euclid-faithful route through XI.11 and XI.8; it is not a claim that the abstract perpendicular-existence statement intrinsically requires Euclidean parallelism.